R16

Code No: 135CZ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
RENEWABLE ENERGY SOURCES
(Common to CE, EEE, ME, ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Discuss different Renewable sources of energy with special reference to Indian Context. [2]
b) What is a fuel cell? What are its main advantages? [3]
c) Comment on origin and distribution of Geothermal Energy. [2]
d) Explain various design of blades of HAWT and their relative features. [3]
e) What are the factors causing winds? [2]
f) Describe the principle behind the operation MHD generator and mention the advantages of MHD system. [3]
g) What do you understand by Energy Farming? [2]
h) Describe the various methods of storage of hydrogen. [3]
i) List the general principles of Energy Conservation. [2]
j) What are the environmental concerns of geo thermal energy? [3]

PART - B

(50 Marks)

2.a) Explain in brief about various types of Geothermal Resources.
b) What are the main problems associated with the usage of salt hydrates as Phase Change materials. [5+5]

OR

3. Define beam, global and diffused radiation. Derive an expression for total Radiation on inclined surface. Show that horizontal surface receives no ground reflected radiation. [10]

4. Describe the Thermal Energy storage of a solar system. [10]

OR

5. What is the present state of Development in Fuel cells? Comment on Environmental effects of Fuel cells. [10]

6. With the help of a schematic diagram, explain the operation of close cycle MHD generating system. [10]

OR

7. Describe the various characteristics and applications of wind turbines. [10]

used paper 16.12.2018PM

8. Calculate the volume of cow-dung based bio-gas plant required for cooking needs of a family of five adults and lightening needs with two 100 CP lamps for three Hours daily. Also calculate the required number of cows to feed the plant. Assume the standard values of data where required. [10]

OR

9. Calculate the volume of Fixed-dome type Bio gas digester for the output of two cows. Also calculate the thermal power available from biogas. Use the following data:
Retention time = 40 days
Dry matter produced = 2 Kg/day/cow
Biogas Yield = 0.22 m^3/kg of dry matter
Percentage of Dry matter in cow dung is 18%
Density of slurry is 1090 kg/m^3
Burner Efficiency is 60% heating value of Biogas is 23 MJ/m^3. [10]

10.a) A deep ocean wave of 2.5 m peak to peak appears at a period of 10 s. Find the wavelength , phase velocity and power associated with the wave. At this power rate, what is the average annual wave energy in MWh/m.

b) Explain the present status of development of Ocean energy resources. [5+5]

OR

11. A single basin type tidal power plant has a basin area of 2 km^2. The tide has an average range of 13 m. Power is generated only during the ebb cycle. The turbine stops working when the head on it falls below 3 m. calculate the average power generated by the plant in single emptying process of the basin if the turbine generator efficiency is 0.7. Estimate the average annual energy generation of the plant. [10]

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Use of paper for 12-2018PM